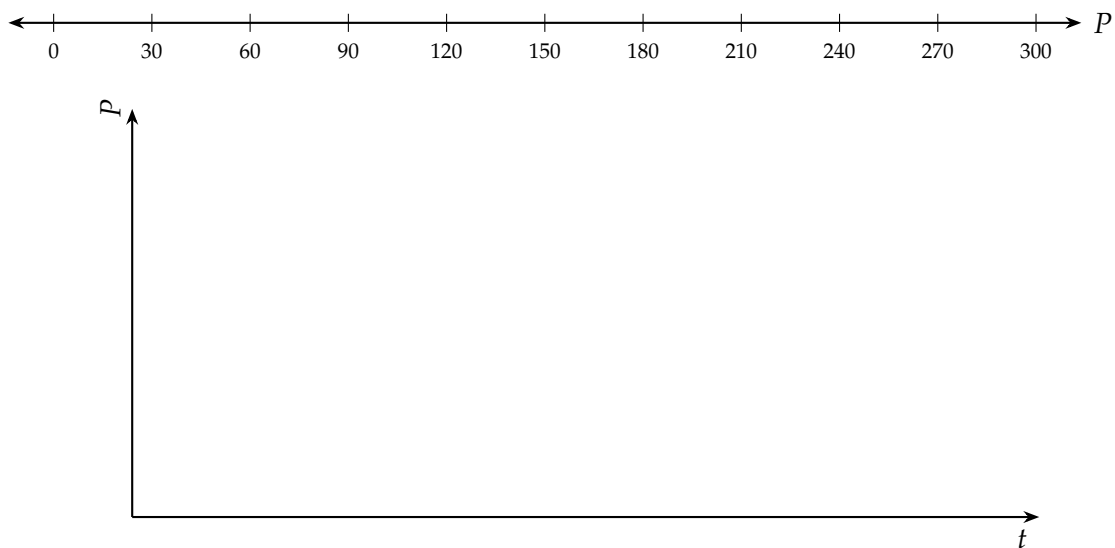


Lecture 16 Live Problems

The shapes of equilibrium. Two state variables.

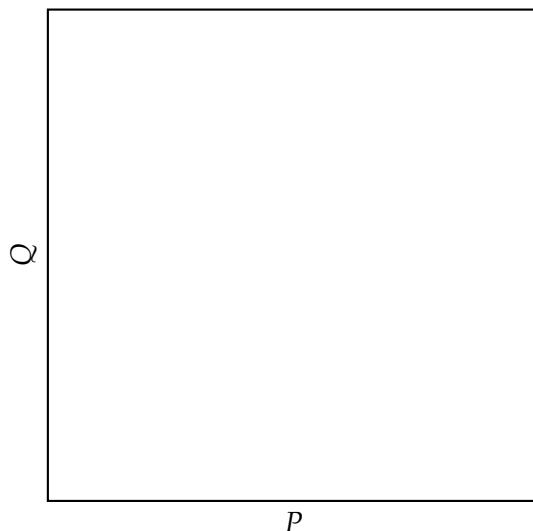
Points toward lab manual pages: **GLV1, GLV2, GLV3, GLV4** (section L3 S5)

1. What is the phase portrait for the change equation $P' = 0.6 P (1 - P/230)$? What do you expect for the long-term behavior? Draw a few time series.



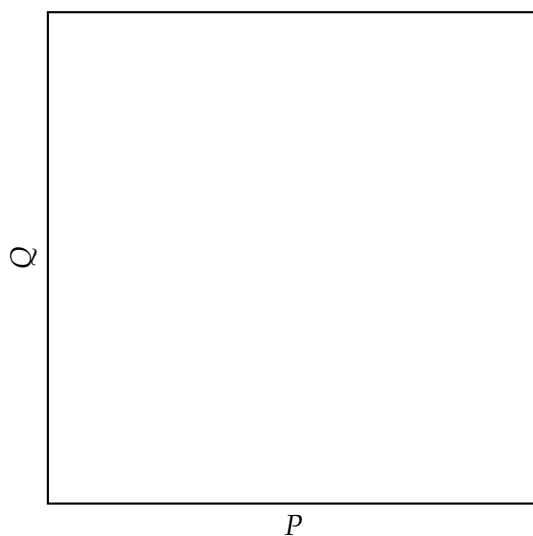
2. Nullclines, no interaction. Plot the nullclines for the change equations

$$P' = b P (1 - P/230), \quad Q' = c Q (1 - Q/300).$$



3. Nullclines, with competition. Describe and plot the nullclines for the change equations

$$P' = P (1 - P/230) - 0.1 P Q, \quad Q' = 2 Q (1 - Q/300) - 0.2 P Q.$$



4. Near the central equilibrium. Analyze the behavior of the system near the central equilibrium point.

